

Detecting Phishing Emails via Textual and Structural Analysis

Project Overview

Developed a machine learning–based phishing email detection system that combines natural language processing (NLP) and structural email analysis to improve phishing classification accuracy. The project was designed to address limitations in traditional phishing detection systems that rely solely on either textual content or structural indicators.

The system analyzes both the linguistic characteristics of emails and structural metadata commonly associated with phishing attacks, enabling more reliable identification of malicious messages while minimizing false positives.

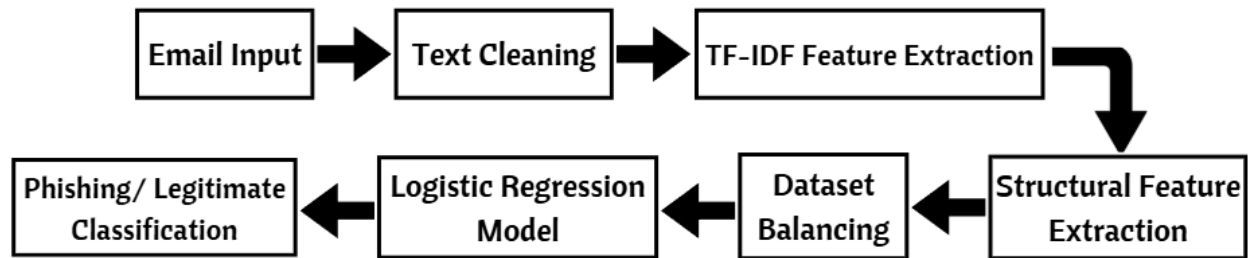
Objectives

- Build an automated phishing email detection pipeline
- Combine textual and structural analysis into a single classification model
- Improve phishing detection accuracy compared to single-feature approaches
- Evaluate the effectiveness of oversampling for handling class imbalance
- Generate reproducible experimental results and evaluation metrics

Technologies and Tools Used

- Python
- Scikit-learn
- Pandas
- NumPy
- Matplotlib
- Natural Language Processing (NLP)
- TF-IDF Vectorization
- Logistic Regression
- Random Oversampling
- VS Code

Workflow Diagram



Dataset and Data Processing

The project utilized multiple publicly available phishing email datasets alongside manually selected legitimate email samples derived from a large corporate email dataset.

Data Processing Pipeline

- Parsed and standardized raw email data
- Extracted email subject lines and body content
- Removed irrelevant characters and normalized text
- Combined email fields into a unified text representation
- Generated structured CSV datasets for model training

Structural Features Extracted

- URL count within email body
- Email body length
- Subject line length

Textual Features Extracted

- TF-IDF vectorized text features
- Tokenized and normalized email content

Machine Learning Methodology

The detection pipeline combined TF-IDF textual vectors with engineered structural features to create a unified feature set for classification.

Model Workflow

1. Dataset preprocessing and cleaning

2. Feature extraction (textual + structural)
3. Dataset balancing using Random Oversampling
4. Model training using Logistic Regression
5. Performance evaluation using classification metrics

An 80/20 stratified train-test split was used to preserve class distribution consistency during evaluation.

Results and Evaluation

The final model achieved extremely high classification performance after balancing the dataset and integrating both textual and structural features.

Performance Metrics

- Accuracy: 99.94%
- Strong precision and recall across both phishing and legitimate email classes
- Minimal false positives and false negatives observed in confusion matrix analysis

Key Findings

- Combining textual and structural analysis significantly improved phishing detection capability
- Oversampling effectively mitigated severe dataset imbalance
- Structural metadata enhanced detection of phishing attempts that used realistic language patterns

Security and Technical Skills Demonstrated

- Cybersecurity threat analysis
- Machine learning model development
- Natural language processing
- Feature engineering
- Data preprocessing and cleaning
- Model evaluation and validation
- Dataset balancing techniques
- Python scripting and automation
- Security-focused research methodology

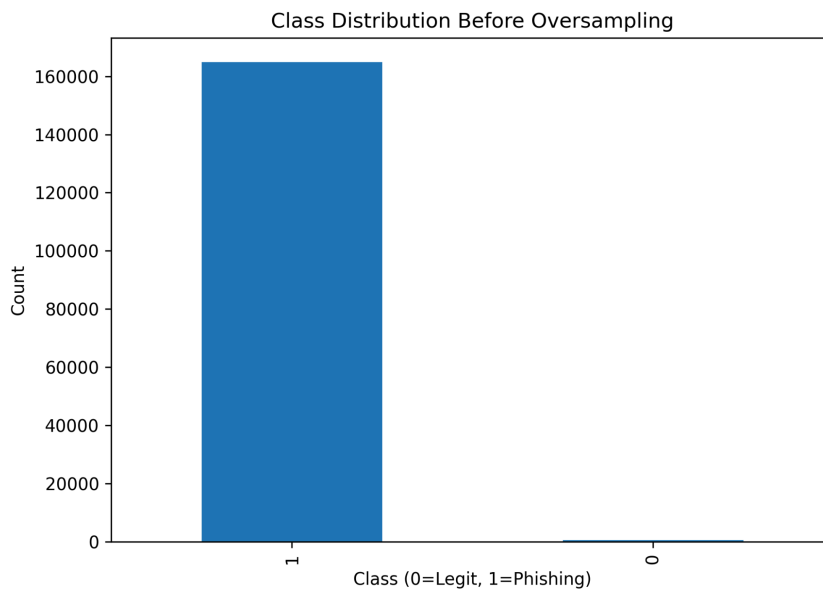
Future Improvements

Potential future enhancements include:

- Real-time phishing detection integration
- Additional structural feature extraction
- Deep learning model experimentation
- Deployment through a web interface or API
- Testing against larger enterprise-scale datasets

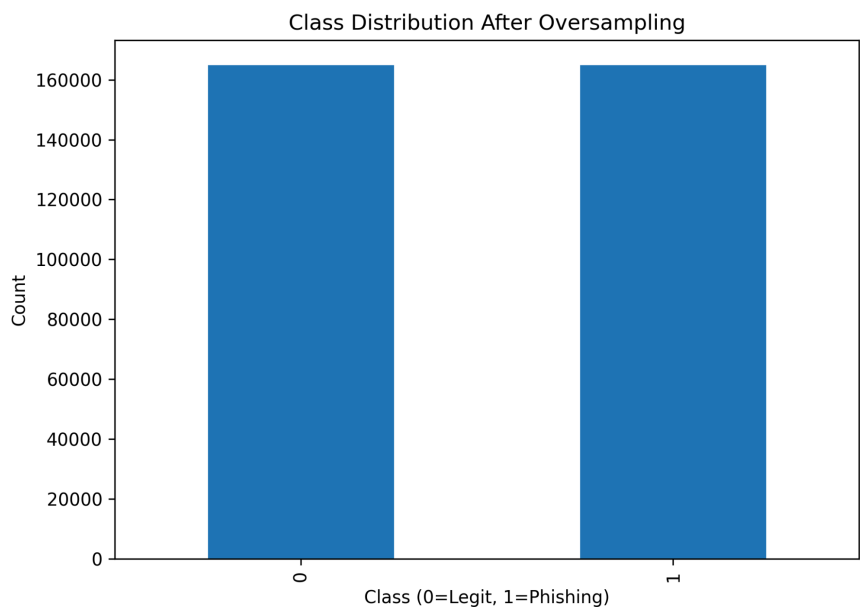
Project Visualizations

Class Distribution Before Oversampling



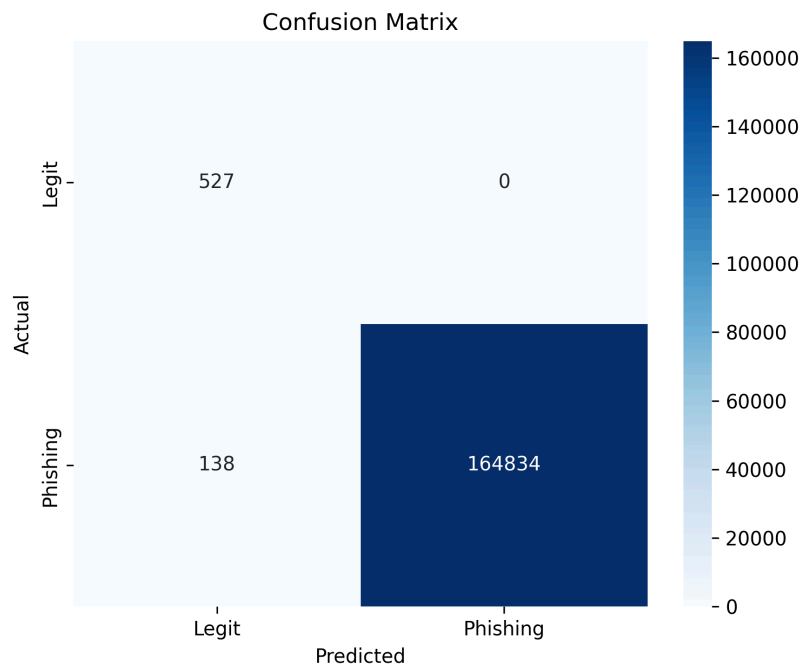
The original dataset contained a significant imbalance between phishing and legitimate emails, which could bias model predictions toward the majority class.

Class Distribution After Oversampling



Random Oversampling balanced both classes, improving the model’s ability to correctly classify legitimate emails and reducing classification bias.

Confusion Matrix



The confusion matrix demonstrates strong separation between phishing and legitimate email classes, with minimal false positives and false negatives.